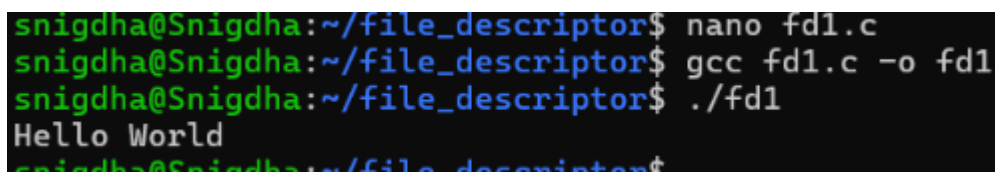


Program 1: stdout — Standard Output

```
#include <unistd.h>

int main()
{
    write(STDOUT_FILENO, "Hello World\n", 12);

    return 0;
}
```

A terminal window showing the execution of Program 1. The user is in the directory ~/file_descriptor. They run 'nano fd1.c' to create the file, 'gcc fd1.c -o fd1' to compile it, and './fd1' to run it. The output is 'Hello World'.

```
snigdha@Snigdha:~/file_descriptor$ nano fd1.c
snigdha@Snigdha:~/file_descriptor$ gcc fd1.c -o fd1
snigdha@Snigdha:~/file_descriptor$ ./fd1
Hello World
snigdha@Snigdha:~/file_descriptor$
```

Program 2: stdin — Standard Input

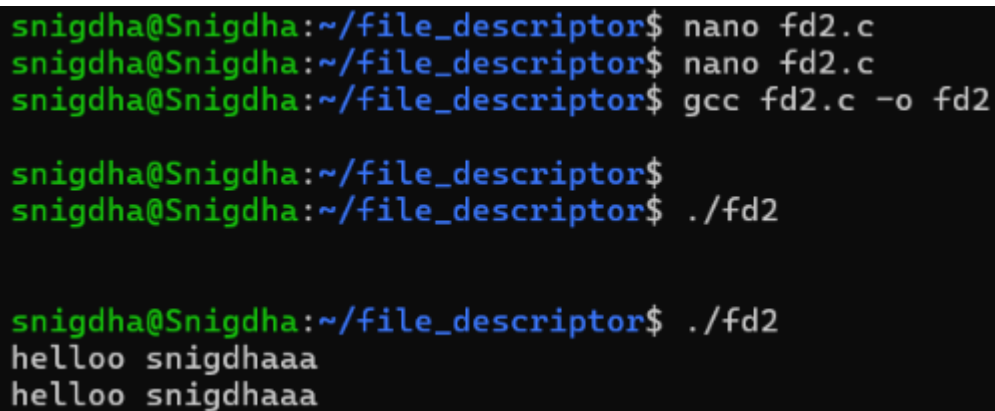
```
#include <unistd.h>

int main()
{
    char buffer[20];

    read(STDIN_FILENO, buffer, sizeof(buffer));

    write(STDOUT_FILENO, buffer, sizeof(buffer));

    return 0;
}
```

A terminal window showing the execution of Program 2. The user is in the directory ~/file_descriptor. They run 'nano fd2.c' to create the file, 'gcc fd2.c -o fd2' to compile it, and './fd2' to run it. They then enter 'helloo snigdhaaa' at the prompt, and the program outputs the same text.

```
snigdha@Snigdha:~/file_descriptor$ nano fd2.c
snigdha@Snigdha:~/file_descriptor$ nano fd2.c
snigdha@Snigdha:~/file_descriptor$ gcc fd2.c -o fd2

snigdha@Snigdha:~/file_descriptor$
snigdha@Snigdha:~/file_descriptor$ ./fd2

snigdha@Snigdha:~/file_descriptor$ ./fd2
helloo snigdhaaa
helloo snigdhaaa
snigdha@Snigdha:~/file_descriptor$
```

Program 3: Proper stdin + stdout

```
#include <stdio.h>

#include <unistd.h>

int main()
{
    char buffer[100];

    int n;

    printf("Enter something: ");

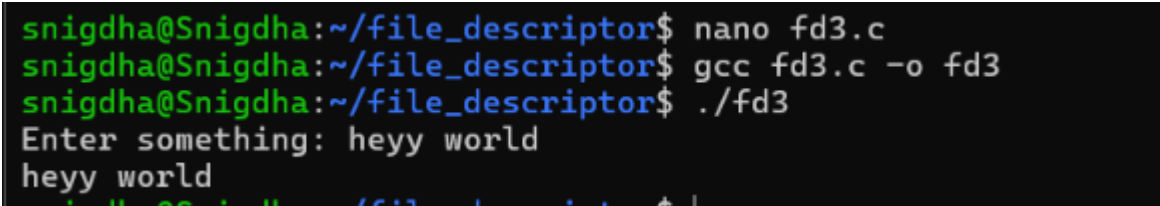
    fflush(stdout);

    n = read(STDIN_FILENO, buffer, sizeof(buffer) - 1);

    if (n > 0)
    {
        buffer[n] = '\0';

        write(STDOUT_FILENO, buffer, n);
    }

    return 0;
}
```



```
snigdha@Snigdha:~/file_descriptor$ nano fd3.c
snigdha@Snigdha:~/file_descriptor$ gcc fd3.c -o fd3
snigdha@Snigdha:~/file_descriptor$ ./fd3
Enter something: hey world
hey world
```

Program 4: stderr — Standard Error

```
#include <unistd.h>

int main()
{
    write(STDERR_FILENO, "This is an error\n", 17);

    return 0;
}
```

```
}
```

As STDERR_FILENO is: 2

write(STDERR_FILENO, "This is an error\n", 17) is equivalent to:

write(2, "This is an error\n", 17);

```
snigdha@Snigdha:~/file_descriptor$ nano fd4.c
snigdha@Snigdha:~/file_descriptor$ gcc fd4.c -o fd4
In file included from fd4.c:1:
fd4.c:8:4: error: expected '=', ',', ';', 'asm' or '__attribute__' before nu
meric constant
    8 |   As STDERR_FILENO is: 2
      |   ^~~~~~
```

Program 5: stderr for an Actual Error

```
#include <stdio.h>
```

```
#include <fcntl.h>
```

```
#include <unistd.h>
```

```
int main()
```

```
{
```

```
    int fd;
```

```
    fd = open("doesnotexist.txt", O_RDONLY);
```

```
    if (fd == -1)
```

```
    {
```

```
        perror("open");
```

```
        return 1;
```

```
    }
```

```
    close(fd);
```

```
    return 0;
```

```
}
```

```
snigdha@Snigdha:~/file_descriptor$ nano fd5.c
snigdha@Snigdha:~/file_descriptor$ gcc fd5.c -o fd5
snigdha@Snigdha:~/file_descriptor$ ./fd5
open: No such file or directory
```

Program 6 Creating file using file descriptor

```
#include <stdio.h>

#include <fcntl.h>

#include <unistd.h>

int main()
{
    int fd;

    fd = open("student.txt", O_CREAT | O_WRONLY, 0644);

    if (fd == -1)
    {
        perror("open");

        return 1;
    }

    printf("File created successfully!\n");

    printf("File descriptor = %d\n", fd);

    close(fd);

    return 0;
}
```

```
snigdha@Snigdha:~/file_descriptor$ nano fd6.c
snigdha@Snigdha:~/file_descriptor$ gcc fd6.c -o fd6
snigdha@Snigdha:~/file_descriptor$ ./fd6
File created successfully!
File descriptor = 3
```

Program 7

```

#include <stdio.h>

#include <fcntl.h>

#include <unistd.h>

int main()
{
    int fd;

    fd = open("student.txt", O_WRONLY | O_TRUNC);

    if (fd == -1)
    {
        perror("open");

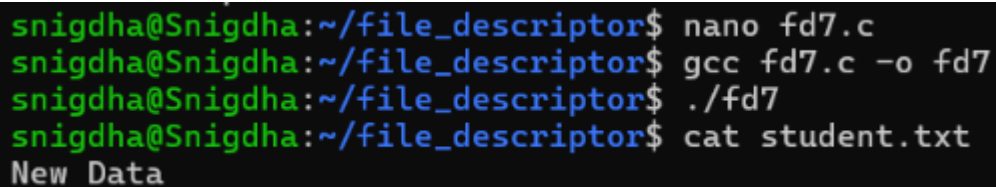
        return 1;
    }

    write(fd, "New Data\n", 9);

    close(fd);

    return 0;
}

```



```

snigdha@Snigdha:~/file_descriptor$ nano fd7.c
snigdha@Snigdha:~/file_descriptor$ gcc fd7.c -o fd7
snigdha@Snigdha:~/file_descriptor$ ./fd7
snigdha@Snigdha:~/file_descriptor$ cat student.txt
New Data

```

Program 8 - Understand dup()

```

#include <stdio.h>

#include <fcntl.h>

#include <unistd.h>

int main()
{
    int fd1, fd2;

    fd1 = open("student.txt", O_CREAT | O_WRONLY, 0644);

```

```
if (fd1 == -1)
{
    perror("open");
    return 1;
}
printf("fd1 = %d\n", fd1);
fd2 = dup(fd1);
printf("fd2 = %d\n", fd2);
close(fd1);
close(fd2);
return 0;
}
```

```
snigdha@Snigdha:~/file_descriptor$ nano fd8.c
snigdha@Snigdha:~/file_descriptor$ gcc fd8.c -o fd8
snigdha@Snigdha:~/file_descriptor$ ./fd8
fd1 = 3
fd2 = 4
```